

1 Introduction

Project Background

This proposal is submitted in response to the tender request by Monash Embedded System Solution (MESS). The objective of this project is to evaluate a high reliability power supply module to give a constant and clean 3.3 V output voltage to a STM32L432K microcontroller. Since power supply is the foundation of the entire system performance, this design focuses on delivering a precise voltage output while ensuring long term hardware safety and noise immunity.

Design Challenges

The STM32 microcontroller is a sensitive component that requires a stable power rail between 1.71 and 4.0 V with a nominal at 3.3 V. However, the system must operate from a variable input source that can reach up to ± 16 V and still maintain a constant output voltage. Additionally, the microcontroller operates under dynamic load condition, with switching clock frequency between 1 MHz and 80 MHz. These different states introduce rapid changes in current demand and generate high frequency noise which affect stability of the output voltage. The primary challenge was to design a regulation circuit that could mitigate power spikes, while preventing noise and ripple to ensure the microcontroller function properly.

Technical objective

To meet the mandatory specifications of the tender, our team identified several core objectives that serve as a checklist for our final design:

- **Regulated Output:** Maintain a constant **3.3 V supply** using **LM317** voltage regulator with precision resistor values. The design must achieve a stable 3.3 V output within the standard **6V to 8V input range**.
- **Voltage Robustness:** Ensures the module remains functional and safe without component burnout when subjected to input voltage extremes up to ± 16 V.
- **Transient Load Stability:** Suppress voltage and current ripple to handle microcontroller **1 MHz and 80 MHz active load switching** to prevent logic reset.
- **Noise Immunity:** Utilize a strategic capacitor network to filter the noise generated by the voltage input and microcontroller clock switching within the mandatory ripple control range.
- **Commercial Viability:** Use **common industrial-grade components** to ensure low cost, reliability and stock availability. The total unit price remains below the **A\$ 5.00** budget for mass production.

Methodology

The development process followed a rigorous engineering workflow. Initial theoretical calculations were done to determine the optimal resistor value and capacitor values. This was followed by LTspice simulations to analyze the circuit behavior under dynamic load conditions. Finally, empirical laboratory testing was conducted by using an oscilloscope to obtain data measurement and a function generator to generate noise in order to verify the prototype's performance against the simulated data. This report presents the details of each stage of this process, and provides technical evidence for our final design.

2 Requirements Analysis

2.1 Project Objective

This project aims to design a power supply for an STM32L432K series microcontroller used in an embedded audio system. The design must develop a stable and low noise power supply circuit under different operational requirements utilizing the LM317 voltage regulator for mass production. The tender project objective is to ensure detailed documentation of the proposed design that satisfies all the client requirements and parameter specifications, including power circuit, protection features, manufacturing purposes, costs constraints and electrical functions.

2.2 Circuit Requirements

2.2.1 Power supply for STM32 microcontroller

- The design **must** deliver a regulated voltage to the STM32L432K series microcontroller within 1.71 V to 4.0 V operating range, with a nominal voltage of 3.3 V. This ensures reliable operation across varying configurations while protecting the microcontroller from damage caused by brownout (undervoltage) and overvoltage conditions [1], [2].
- The output minimum current **should** handle 140 mA for fixed load and 10 mA for active load to power the circuit components.
- The power supplied **must** be consistent with low noise and ripple, acquiring a maximum voltage ripple of 5%, 1MHz active load current ripple of 5% and 80MHz at 10%.
- The proposed design **must** have the ability to be tested and function properly within -16 V to 16 V with a source voltage between 6 V to 8 V into the LM317 regulator.

2.2.2 LM317 Voltage Regulator

This circuit **must** integrate the LM317 voltage regulator to provide a stabilised output at standard 3.3 V. The LM317 regulates output voltage using a fixed internal reference voltage of 1.25 V between the adjacent and output pins [3].

2.2.3 Electrical Circuit

The power supply system **can** use any topology to achieve the client's requirements.

2.3 Protection Requirements

2.3.1 Reverse Polarity Protection

The design **must** prevent damage to the circuit due to incorrect voltage source polarity. This is because both voltage regulator and microcontroller are semiconductor-based devices; reverse bias beyond design limits causes breakdown of PN junctions and unintended current conduction paths, leading to thermal failure [1], [3], [4], [5].

2.3.2 Overvoltage protection

The output voltage flowing to the STM32 load **must** have a nominal 3.3 V value without exceeding a maximum of 4 V. An overvoltage protection is essential for a larger safety margin as overvoltage would permanently damage the circuit components, specifically the microcontroller where its operating voltage ranges from 1.71 V to 3.6 V [1], [2].

2.4 Manufacturing requirements

2.4.1 Eleclab Toolkit components

The demonstration design prototype **must** only be built using components found in the Eleclab Toolkit for data measurement and physical lab testing.

2.4.2 Electrical components

The proposed final design components **may** be from another supplier or package format and it is not mandatory to use components from the Eleclab Toolkit.

2.4.3 Backup Components

All components used in the final proposed design **must** be available from at least two different sources or suppliers with high reliability in the required quantity. This ensures components are readily available for mass production and reduces supply chain disruptions and delays in production.

2.4.4 Lead Time Constraints

Electrical components used **should** have short lead time and shipping cycles to ensure operational efficiency and reduce material shortages.

2.5 Cost requirements

2.5.1 Low Cost

The design **must** fulfil a target cost of \$5 AUD per unit under a production scale of 10,000 units per year. This design **should** prioritise the use of widely available, standard components and avoid customised or specialised parts.

2.5.2 Other Manufacturing Costs

The design **should** minimise overall production costs, which include the choice of boards, assembly costs, and component installations. In addition, the amortised non-recurring engineering (NRE) costs including the research, development, testing, and engineering labour **should** remain within 10% of the total unit cost over a production volume of 10,000 units, ensuring cost-effectiveness.

2.4 Electrical Parameter Specification

Table 1: Electrical Parameter Specifications

No	Parameter specification	Value			Type requirements of	Link to Requirements
		Min	Nominal	Max		
1	Input voltage for STM	1.71V	3.3 V	4.0 V	Power supply circuit Overvoltage protection	MUST achieve between the required limit and never exceeding maximum value to prevent damage to the microcontroller.
2	DC load current	140 mA			Power supply circuit	Ensure sufficient and stable power for load under steady state.
3	Active load current	10 mA			Power supply circuit	Ensure sufficient and reliable power for transient behaviour of STM32 clocking acting as active load.
4	DC load Voltage Ripple			< 5 %	Power supply circuit	Reduces ripple and noise for consistent output power supply to preserve audio quality.
5	Voltage variance (> 1 minute)		0.1 V	0.2 V	Power supply circuit	Ensure stability for a long time at different loads.
6	1 MHz current ripple			< 5 %	Power supply circuit	Ensure current remain stable during STM32 clocking demonstrated using mosfet
7	80 MHz current ripple			< 10 %	Power supply circuit	Ensure current remain stable during high frequency STM32 clocking demonstrated using mosfet
8	Input voltage		6 V	8 V	Overvoltage protection	Limit voltage into the LM317 regulator to prevent damage
9	Test voltage		-16 V	16 V	Power supply circuit	Protect circuit component and ensure safety of operator when testing

3 Design Process

3.1 Assumptions

- 1) This design assumes an ideal circuit model where there is no internal resistance in voltage sources, no wire resistance and no leakage current. These effects are considered negligible due to the low current operation and short connection lengths in circuit.
- 2) The design would operate between -16 V to 16 V with limited current and would never go beyond this range.
- 3) Analytical calculations assume ideal resistor, capacitor, zener diode, diode and MOSFET.
- 4) Environmental effects like temperature changes, humidity, electromagnetic interference and corrosion are not taken into consideration.
- 5) The design uses only components from the Eleclab Toolkit for prototyping and data measurement.

3.2 Design Process

3.2.1 Input Voltage for STM32 Using LM317 Voltage Regulator

The first stage of the design process was to produce a nominal 3.3 V output voltage that acts as a power supply for STM32 regulated using the mandatory LM317 voltage regulator. LM317 was selected as it is a compulsory requirement and provides an adjustable voltage using different resistor values. It regulates voltage using a fixed internal reference voltage of 1.25 V between the output and adjacent terminals. Thus, the desired output voltage could be adjusted using Equation 1 specified in [3]:

$$V_{out} = V_{ref} * (1 + \frac{R2}{R1}) \quad (1)$$

Table 2: Design Conditions

Parameter	V_{in}	V_{out}	V_{ref}	I_{min_load}
Value	-16 V to 16 V	3.3 V	1.25 V	3.5 mA

A standard resistor value of $R1 = 220 \Omega$ was chosen due to its common availability, whereas $R2$ value was calculated using Equation 2:

$$R2 = (\frac{V_{out}}{V_{ref}} - 1) * R1 = (\frac{3.3}{1.25} - 1) * 220 = 360.8 \Omega \quad (2)$$

Since 360.8Ω is not a standard resistor value available, a practical value of 364Ω was implemented using a series combination of standard E3 resistors. Substituting $R1 = 220 \Omega$ and $R2 = 364 \Omega$ into Equation (1) yields a V_{out} of 3.318 V (see Appendix B1).

Thus, V_{out} has satisfied the targeted 1.71 V to 4.0 V operating range requirement, despite a slight deviation of 0.018 V above the nominal 3.3 V . The minimum load current required by LM317 was checked using Equation 3 below:

$$I_{min_load} = \frac{V_{out}}{R1 + R2} = \frac{3.318}{220 + 364} = 5.68 \text{ mA} \quad (3)$$

The current value satisfies the typical minimum load current of 3.5 mA mentioned in the LM317 datasheet, approving the $R1$ and $R2$ resistor value choices. The proposed design using LM317 is as shown below:

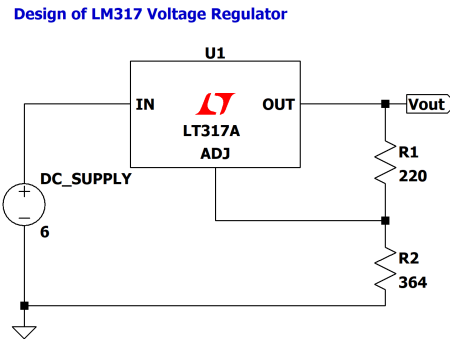


Figure 1: Schematic of LM317 voltage regulator with chosen $R1$ and $R2$ resistor values

3.2.2 Passive and Active Load Output Current

To verify that the integrated circuit can satisfy the minimum load output current requirement, the STM32 microcontroller was modelled using a load resistor $R3$ and a VN2222LL-G (TO-92 package) Field Effect Transistor (FET).

For fixed DC load conditions, the MOSFET was set to output a constant voltage source of 5 V , higher than the gate source voltage. Using the nominal output voltage of 3.318 V and a targeted minimum current of 140 mA , the theoretical load resistance is:

$$R_{DC} = \frac{V}{I} = \frac{3.318 \text{ V}}{0.14 \text{ A}} = 23.7 \Omega \quad (4)$$

A practical resistor of 10Ω was selected as it provides a minimum 140 mA output current using E3 resistor value (see Appendix B2). This value was used to verify that the voltage regulator can provide a stable output voltage under steady current demand.

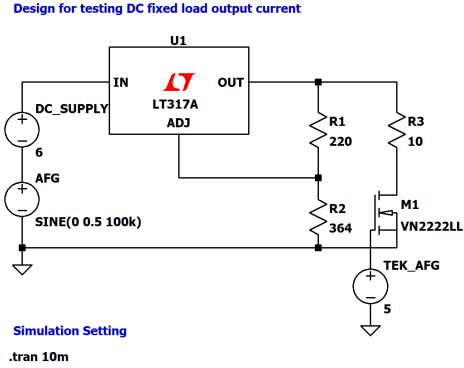


Figure 2: Schematic of a DC load using resistor R3 and a constantly on MOSFET M1. Active load conditions were simulated by using a 80 MHz high frequency square wave pulse to continuously switch the MOSFET on and off, as shown in Figure 2. Using targeted minimum current of 10 mA, the theoretical load resistance is:

$$R_{AC} = \frac{V}{I} = \frac{3.318 V}{0.01 A} = 331.8 \Omega \quad (5)$$

A practical resistor of 220 Ω was selected as it provides a minimum 10mA output current using E3 resistor value (see Appendix B3). Another shunt resistor R4 of 220 Ω was connected in parallel with the load to measure the output current, as shown in Figure 3.

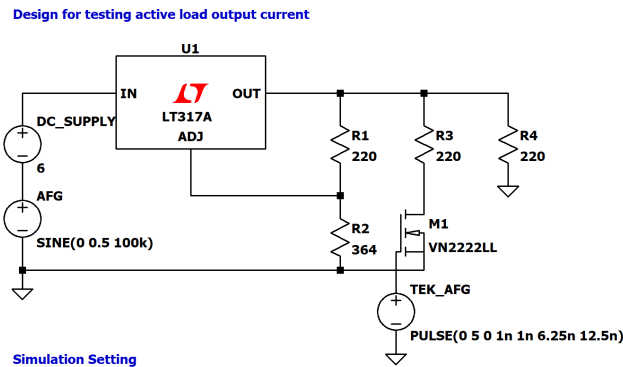


Figure 3: Schematic of an active load using resistor R4 and a 80 MHz MOSFET M1

3.2.3 Voltage and Current Ripple Reduction under Active Load

The next design stage was to reduce output voltage and current ripple to provide a stable power supply for STM32. Voltage and current ripple was considered to come from two main sources. The first source was low noise input, represented using a 100kHz sinusoidal AC component by a function generator superimposed on a DC voltage level. The second source was active load that results in 80 MHz high frequency transient noise, represented using an actively switching mosfet that mimicked the behaviour of the STM32.

The percentage ripple was calculated using Equation 6 and Equation 7:

$$\text{Voltage Ripple \%} = \frac{V_{\text{peak-to-peak}} / V_{\text{ripple}}}{V_{\text{out active load}}} \times 100 \quad (6)$$

$$\text{Current Ripple \%} = \frac{I_{\text{peak-to-peak}} / I_{\text{ripple}}}{I_{\text{mean}}} \times 100 \quad (7)$$

It is observed that under a stabilised 6 V input voltage, the voltage ripple is 11.181 % and the current ripple is 12.170%, which does not meet the client's requirements. To smooth out the output voltage and current ripple, the first filtering capacitor was placed from the regulator adjacent pins to ground with a connection between R1 and R2 resistors. This resistor serves to reduce high frequency input to the ground and prevent amplified by LM317.

Table 3: Ripple percentage after adding capacitor 1

Capacitor 1 Value	V _{OUT} (V)	I _{R4} (mA)	V _{RIPPLE ACTIVE} (mV)	I _{RIPPLE ACTIVE} (mA)	Voltage Ripple ACTIVE (%)	Current Ripple ACTIVE (%)
C1 = 0.1 µF	3.314	14.938	352.945	1.788	10.650	11.969
C1 = 1 µF	3.345	15.213	290	1.573	8.670	10.340
C1 = 10 µF	3.357	15.288	266.384	1.536	7.935	10.047

After evaluating three different capacitor values as shown in Table 3, the 1µF capacitor was chosen as it provides a significant reduction in voltage ripple from 11.181% to 8.670%, and reduces current ripple from 12.170% to 10.340% while maintaining cost-effectiveness. Although the 10µF capacitor has better ripple reduction performance, the additional improvement was not sufficient to justify the increased cost.

A second capacitor was added from the LM317 regulator input to the ground. This capacitor works to reduce input voltage and current ripple before entering the LM317 regulator affected by both the noisy input and transient noise travelling back to the input side.

Table 4: Ripple percentage after adding capacitor 2

Capacitor 2 Values	V _{OUT} (V)	I _{R4} (mA)	V _{RIPPLE ACTIVE} (mV)	I _{RIPPLE ACTIVE} (mA)	Voltage Ripple ACTIVE (%)	Current Ripple ACTIVE (%)
C1 = 1 µF C2 = 1 µF	3.331	14.926	275.103	1.467	8.259	9.828
C1 = 1 µF C2 = 10 µF	3.304	14.832	254.692	1.354	7.709	9.071
C1 = 1 µF C2 = 47 µF	3.307	14.810	186.816	1.301	5.649	8.785

Following the selection of $C1 = 1\mu\text{F}$ from Table 4, the second capacitor was chosen to be $47\mu\text{F}$. This capacitor results in a further deduction in voltage ripple from 8.670% to 5.649%, and current ripple from 10.340% to 8.785%. The significant reduction in voltage ripple justified the increase in cost in choosing capacitors with larger values, while improving circuit stability and preventing electric noise.

The third capacitor was added from the regulator output to the ground. The purpose of this capacitor is to reduce noise introduced by the output active loads which propagate to the input side of the voltage regulator and destabilise the circuit.

Table 5: Ripple percentage after adding capacitor 3

Capacitor 3 Values	V_{OUT} (V)	I_{R4} (mA)	$V_{\text{RIPPLE ACTIVE}}$ (mV)	$I_{\text{RIPPLE ACTIVE}}$ (mA)	Voltage Ripple ACTIVE (%)	Current Ripple ACTIVE (%)
$C1 = 1\mu\text{F}$ $C2 = 47\mu\text{F}$ $C3 = 0.1\mu\text{F}$	3.388	15.406	77.378	0.352	2.284	2.284
$C1 = 1\mu\text{F}$ $C2 = 47\mu\text{F}$ $C3 = 1\mu\text{F}$	3.391	15.394	51.821	0.237	1.528	1.540
$C1 = 1\mu\text{F}$ $C2 = 47\mu\text{F}$ $C3 = 10\mu\text{F}$	3.392	15.416	10.526	0.046	0.310	0.298

After fixing $C1 = 1\mu\text{F}$ and $C2 = 47\mu\text{F}$, the third capacitor was chosen to be $1\mu\text{F}$. The voltage ripple was reduced from 5.649% to 1.528% and current ripple from 8.785% to 1.540%. While having a $10\mu\text{F}$ capacitor could significantly achieve a lower ripple value, the additional cost needed is not justified as the design requirements were already satisfied.

Therefore, the final selected capacitor values are $C1 = 1\mu\text{F}$, $C2 = 47\mu\text{F}$ and $C3 = 1\mu\text{F}$, achieving a final voltage ripple of 1.528% and current ripple of 1.540% (see Appendix B4), successfully meeting the client's requirements of having maximum 10% current ripple under 80MHz active load switching. Under a MOSFET switching under 1 MHz, the active load current ripple is 1.155%, which also satisfies the requirement of being under 5% (see Appendix B5).

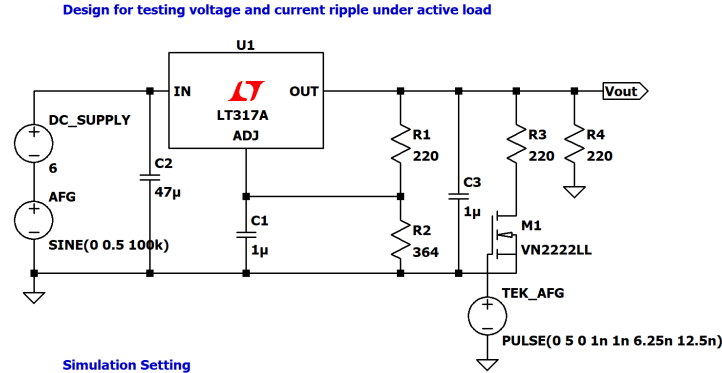


Figure 4: Schematic of testing current ripple under active load

3.2.4 Voltage and Current Ripple Reduction under DC Fixed Load

When all three capacitors are added as shown in Figure 5, the circuit was tested under a DC fixed load condition, where the MOSFET was continuously turned on by providing a gate-source voltage of 5V. The voltage ripple was found to be 1.496% and current ripple to be 1.503% (see Appendix B6). These results successfully meet the design requirements of maintaining DC voltage ripple to be under 5% where the circuit is now only influenced by the 100kHz input noise.

Table 6: Ripple percentage after adding capacitor 3

Capacitor 3 Values	V_{OUT} (V)	I_{R3} (mA)	$V_{RIPPLE\ DC}$ (mV)	$I_{RIPPLE\ DC}$ (mA)	Voltage Ripple _{DC} (%)	Current Ripple _{DC} (%)
C1 = 47 μ F C2 = 1 μ F C3 = 1 μ F	3.389	238.743	50.686	3.588	1.496	1.503

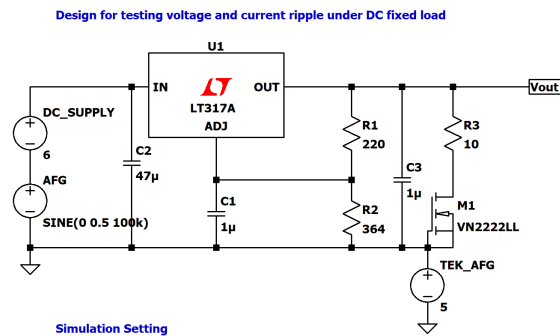


Figure 5: Schematic of testing voltage ripple under DC fixed load

3.2.5 Overvoltage Protection

The next design stage was to ensure that the voltage entering LM317 never exceeds its safe operating range. This overvoltage protection approach used is by zener diode clamping. A 3.6 V

zener diode was added after the voltage regulator in parallel. This is to ensure that the output voltage to STM32 would be kept under 3.6 V and strictly never exceed the maximum 4 V, as per the requirements. The zener diode protects the microcontroller from potential damage caused by faulty LM317 or incorrect resistor values that result in sudden voltage spikes.

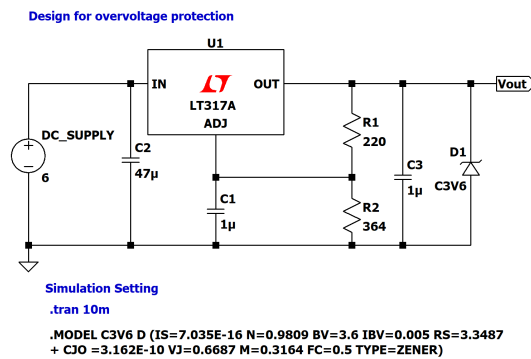


Figure 6: Schematic of overvoltage protection using 3.6V zener diode D1

3.2.6 Reversed Polarity Protection

The design was required to protect the circuit against reversed polarity of the voltage source by utilizing the diode behaviour of allowing current to flow in one direction. A diode is placed at the input of the voltage regulator to prevent damage to LM317 from misconnection during operation. When the polarity is incorrect, current is blocked and no voltage would be supplied to the voltage regulator (see Appendix B7).

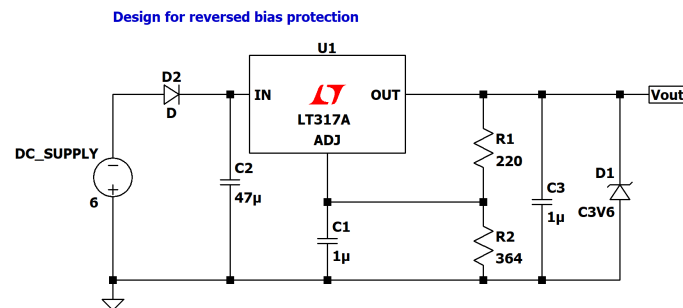


Figure 7: Schematic of reversed polarity protection using 1N4007 diode D2

3.3 Justification

3.3.1 Justification of LM317 Regulator Output Voltage

LM317 voltage regulator is chosen as it is a mandatory requirement and an effective way to regulate voltage based on different values of the resistor chosen. The resistor values uses standard E3 resistors instead of the ideal calculated values as the design values low cost and focuses on mass production. Thus, using a series combination of standard resistor to achieve 364 Ω allows output voltage to remain at 3.318 V with only a slight deviation from the nominal 3.3 V while being cost effective and easy to source from different suppliers.

3.3.2 Justification of Passive and Active Load Output Current

Data measurement and testing on both passive and active load is justified as the microcontroller would not behave as a single constant resistor in real life operations. The fixed DC load represents the power supply's ability to provide the minimum current for passive load, and active load for STM32 rapid clock switching. Thus, it is necessary to perform both measurements to access the output voltage supplied.

The use of MOSFET to represent the behaviour of STM32 microcontroller switching is also justified as it provides a flexible way to test the power supply under different load conditions. MOSFET can be adjusted by providing either a constant voltage source for passive load or a square wave for active load, which makes it suitable for testing the design.

3.3.3 Justification of Voltage and Current Ripple Reduction

The three capacitors are added in multiple locations across the circuit to filter noise and reduce ripple effect on output voltage and current caused by input disturbance and active load switching. After filtering, the output voltage ripple under DC load was found to be 1.496%, current ripple under 1 MHz is 1.155% and 1.540% for under 80 MHz active load. The design process uses trial-and-error methods on different capacitor values to determine the best combination. Standard E3 series electrolytic capacitors are used to provide sufficient ripple reduction with minimal cost.

3.3.4 Justification of Overvoltage Protection

The overvoltage protection design is justified as the requirements state that output voltage flowing to STM32 should not exceed 4 V at any time. Thus, a 3.6 V zener diode is connected to the output of the LM317. If there appears to be an overvoltage condition such as LM317 fails to work, incorrect resistor selection or voltage spikes, the zener diode can clamp the output voltage at 3.6 V, remaining safely under the maximum 4 V.

An alternative considered was to construct a voltage divider circuit. However, this was rejected as it cannot maintain a fixed voltage when the input voltage fluctuates or under varying loads. Thus, it is highly inefficient and wastes power compared to zener diodes.

3.3.5 Justification of Reversed Bias Protection

The use of a diode in reverse bias protection is justified by its cost effectiveness and ease of implementation. By reversing both the input voltage sources, it ensures no current flows and no voltage is applied to the voltage regulator to prevent overheating and regulator failure.

4 Proposed Design

4.1 Prototype Overview

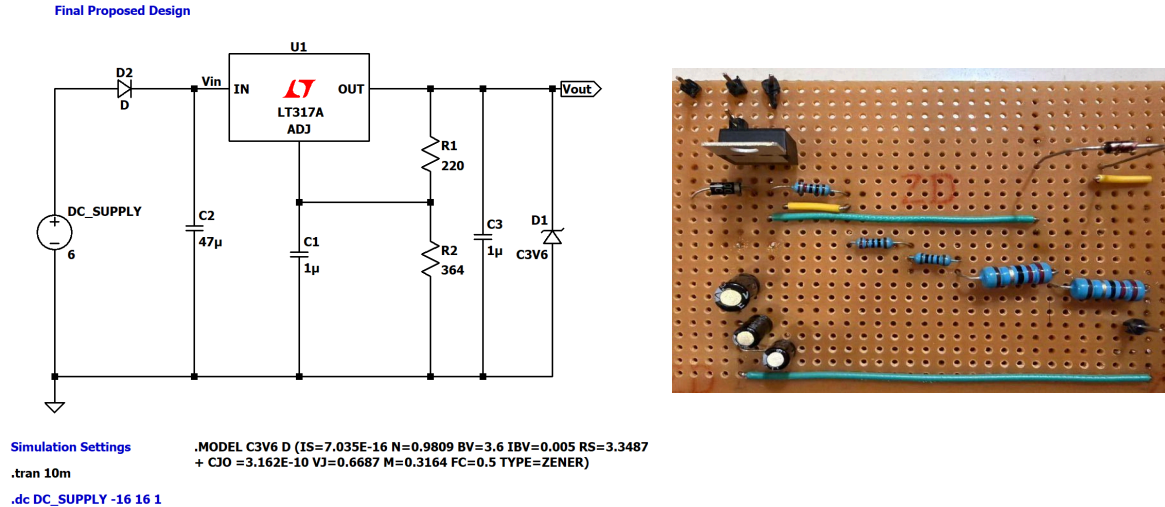


Figure 8: Schematic of proposed design (left) and physical prototype on a veroboard (right)

On the soldered veroboard, the voltage regulator is located at the edge to allow effective heat dissipation. The components are not too close to each other to allow enough spacing for soldering iron tips, preventing solder bridging that results in short circuit. In addition, three probe points are provided at the ground, output and input nodes for easy connection to equipment during testing.

4.1.1 Output voltage using LM317

The design functions to provide a nominal 3.3 V output voltage to the STM32 microcontroller using the LM317 voltage regulator. LM317 is chosen not only as a mandatory requirement, but it is also highly reliable and commonly used in industrial settings. The circuit is constructed using $R1 = 220 \Omega$ and $R2 = 364 \Omega$, which works stably under varying load.

4.1.2 DC Fixed and Active Load Current

Under DC fixed load, the resistor used is 10Ω and the obtained output current is 238 mA. Under active load conditions generated using 80 MHz MOSFET, the resistor used is 220Ω and the obtained current load is 15.4 mA. These results demonstrate that the output current would be sufficient to provide a stable output power to STM22 under varying load conditions.

4.1.3 Voltage and Current Ripple

After smoothing both the input noise and actively switching output load using a total of 3 capacitors of values $C1 = 1\mu\text{F}$, $C2 = 47\mu\text{F}$ and $C3 = 1\mu\text{F}$, the output voltage ripple was found to be 0.545 %, with current ripple of 1.540 % measured at 80 MHz load switching and 1.155 % for 1 MHz. The small ripple observed can significantly reduce voltage fluctuations and improve microcontroller performance in audio recording and processing.

4.1.4 Overvoltage Protection

Overvoltage protection was achieved by adding a 3.6 V zener diode at the output of the voltage regulator. It functions to protect the STM32 microcontroller from damage in case of voltage spikes due to malfunctioning components.

4.1.5 Reversed Bias Protection

The design is protected from incorrect polarity at voltage sources by adding a diode before the input voltage into LM317. The property of the diode would only allow current to flow in the correct direction, thus when reversing the polarity of the voltage source, the output voltage was found to be 0 V, which indicates that the circuit is protected.

4.2 Evaluation Against Requirements

4.2.1 STM Input Voltage

Requirement: The output voltage to power STM32 must range between 1.71 V to 4.0 V with nominal 3.3 V under test conditions between ± 16 V. The output voltage must also provide a stable output when the input voltage is within the normal operating range of 6 V to 8 V.

Proposed Design: The LM317 voltage regulator was chosen as it is a mandatory requirement from the client and provides adjustable voltage configured using a resistor divider network. The resistor values $R1 = 220 \Omega$ and $R2 = 364 \Omega$ were chosen to obtain the closest targeted 3.3 V, while improving cost and availability by using standard E3 series resistors. The output voltage can be calculated using the Equation 8, assuming a 1.25 V internal reference voltage for LM317:

$$V_{out} = V_{ref} * (1 + \frac{R2}{R1}) = 1.25 * (1 + \frac{364}{220}) = 3.318 \text{ V} \quad (8)$$

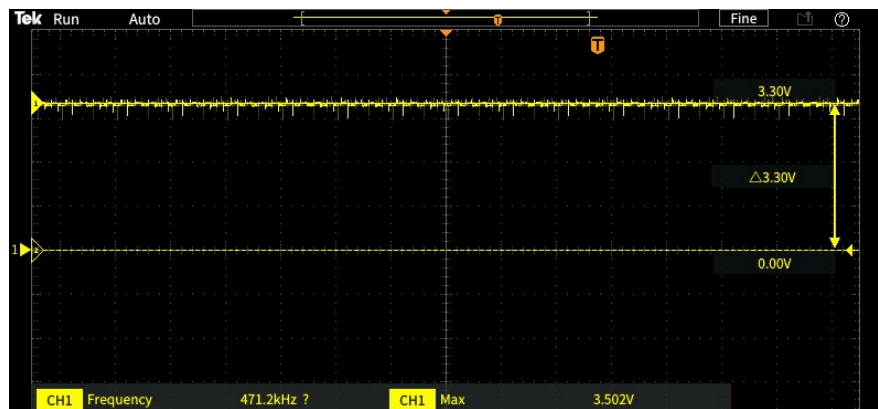


Figure 9: DSO measurement for output voltage

As shown in Figure 9, the measured output voltage stabilises at 3.3 V when the input voltage is greater than 6 V, which successfully meets the required specifications to power STM32 microcontroller. The output voltage lies within the safe operating range of 1.71 V to 4.0 V when tested for input voltage from -16 V to 16 V.

4.2.2 Voltage Ripple under DC Load

Requirement: Under DC load condition, the output voltage ripple should be less than 5 %.

Proposed Design: To measure voltage ripple, a 100 kHz sine wave is generated in addition to the constant DC power supply to represent low noise input. The circuit was tested under fixed load condition using a load resistor of 10 Ω . To reduce output ripple, a series combination of capacitors with values C1 = 47 μ F, C2 = 1 μ F and C3 = 1 μ F were added. Different capacitor values were tested during the design process, and the selected capacitor balances between reducing ripple while being cost effective and offers high manufacturability. These capacitors stabilise the output voltage by storing charges when voltage increases and supplying current when voltage drops.

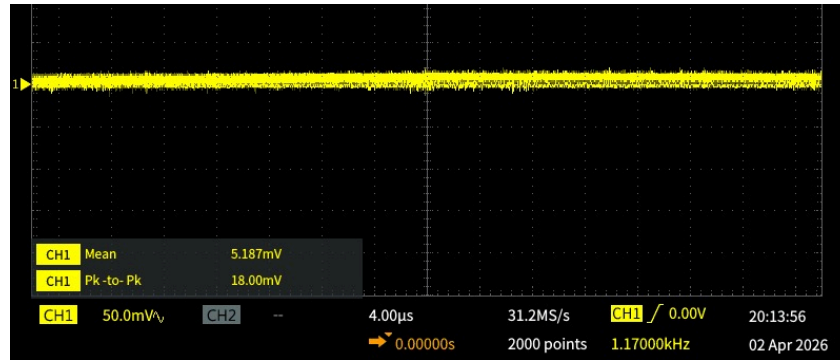


Figure 10: DSO measurement of voltage ripple under DC load using AC coupling mode

According to Figure 10, the peak-to-peak voltage is 18 mV. Using the ripple Equation 9:

$$\text{Ripple \%} = \frac{V_{\text{peak-to-peak}}}{V_{\text{out DC}}} \times 100 = \frac{18 \text{ mV}}{3.3 \text{ V}} \times 100 = 0.545 \% \quad (9)$$

The voltage ripple calculated is 0.545%, which is less than the required 5%. The proposed design successfully reduces output noises as input voltage fluctuates under DC load, indicating that the capacitor effectively filters out noise generated. By adding a diode, the circuit ensures the C1 capacitor directly affects V_{in} of LM317, and not disturbed by voltage sources due to its reversed polarity property, which will significantly reduce voltage and current ripple than what was calculated in the design process.

4.2.3 Current Ripple Under Active Load

Requirement: Under active load condition, the output current ripple should be less than 5% at 1 MHz load switching frequency and less than 10% for 80 MHz switching.

Proposed Design: To simulate the behaviour of an active load, a VN2222LL mosfet was set to switched at 1 MHz or 80 MHz. An additional 220 Ω shunt resistor was connected in parallel after the load to measure current ripple using the same capacitor value. Maintaining a stable output current is important to ensure the STM32 microcontroller receives stable current with little fluctuations caused by high-speed clocking under varying load conditions.

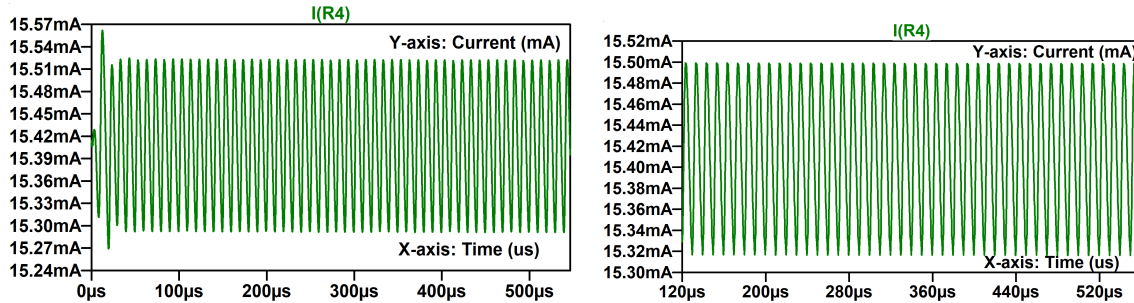


Figure 11: LTspice simulation of current ripple under 80 MHz (left) and 1 MHz (right)

From Figure 11, the output current ripple at 80 MHz was found to be 1.540%, which is below the required maximum 10%. At 1 MHz, the output current was 1.155%, which is also below the required maximum 5%.

4.2.4 Voltage Variance over 1 Minute

Requirement: The voltage variance of the circuit over 1 minute should have a nominal value of 0.1 V and must not exceed maximum 0.2 V.

Proposed Design: From simulation, the output voltage remains constant at 3.391V for over 1 minute, indicating that the circuit reaches a steady state behaviour, as shown in Figure 12.

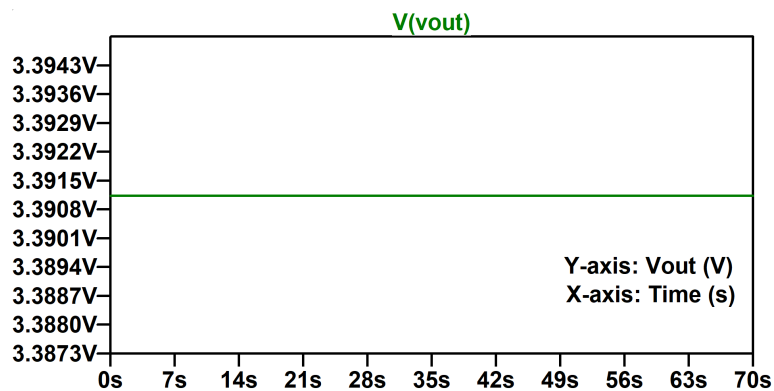


Figure 12: LT spice simulation of output voltage over 1 minute time interval

However, taking into account real world factors such as temperature changes and components tolerance, voltage variance was also measured using a physical circuit. Two sets of data were

recorded at $t = 0\text{s}$ and another at $t = 60\text{s}$ (see Appendix A). The mean voltage was calculated for each dataset, and the voltage variance was calculated using Equation 10:

$$\Delta V = |V_{60} - V_0| = |3.299872 - 3.30142| = 1.548 \text{ mV} \quad (10)$$

The voltage difference between them is 1.548 mV which is well below the maximum 0.2 V, indicating that the actual physical circuit is stable with negligible voltage difference. This results from the LM317 effective voltage regulation with thermal management and capacitor network in maintaining a consistent output voltage.

4.2.5 DC Fixed and Active Output Load Current

Requirement: The power supply for STM32 must provide a minimum output load current of 140 mA under DC fixed load, and minimum of 10 mA under active switching load.

Proposed Design: The STM32 microcontroller was modelled using a series combination of resistor and MOSFET, where the resistor would represent the load current, and the VN2222LL MOSFET to simulate switching behaviour.

DC fixed load

Under DC fixed load condition, the MOSFET is supplied with a constant voltage of 5 V that exceeds the gate source voltage, allowing the MOSFET to be continuously on for current to flow. A practical value of 10Ω was chosen to draw a minimum current of 140 mA.

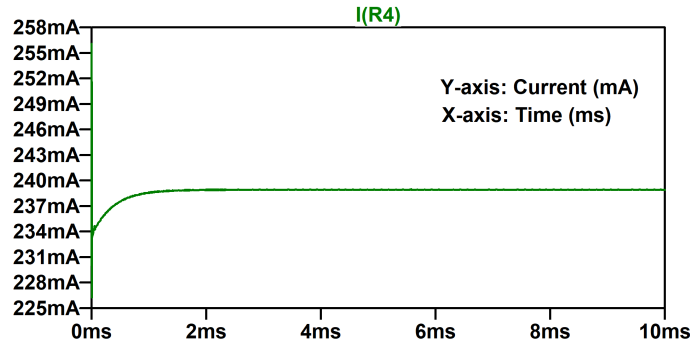


Figure 13: LT spice simulation of output current under DC fixed load

From Figure 13, the measured current remains constant at 238 mA, which satisfies the requirement of achieving minimum load current of 140 mA. The main advantage of the proposed design is its stable thermal management of the LM317 voltage regulator. During operation, the microcontroller constantly draws current while performing background tasks. However, the regulator maintains a stable output voltage and prevents components from heating up over time. By consistently supplying the minimum current needed, the design ensures the STM32 receives sufficient power needed under all operating conditions.

Active Load

For active load, the MOSFET was supplied with a 80 MHz pulse to simulate STM32 high frequency clock switching behaviour. A practical value of 220 Ω was chosen to draw a minimum current of 10 mA.

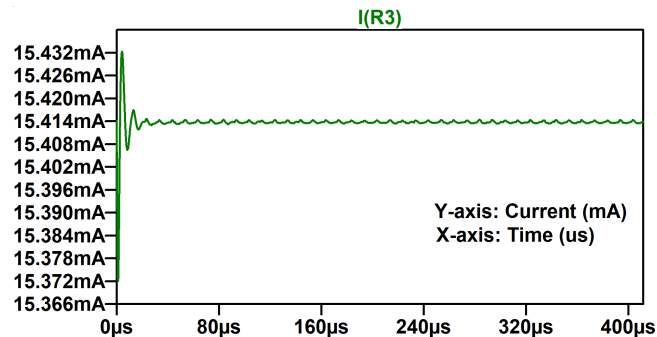


Figure 14: LT spice simulation of output current under 80 MHz active load

Figure 14 shows that the output current achieves the requirements as it remains stable at around 15.4 mA despite high frequency switching. There are no significant voltage drops, thus indicating that LM317 is capable of responding to active load conditions. By having a minimum current above 10 mA, our design ensures that STM32 microcontrollers never enter the brownout state, where the microcontroller would reset due to insufficient power. According to the STM32 datasheet [1], [6] the brownout threshold current is 2.96 mA, while the measured output current is significantly higher.

4.2.6 Overvoltage Protection

Requirement: The output voltage should remain between 1.71 V and 4 V, and must strictly never exceed the maximum values under normal or faulty conditions.

Proposed Design: To provide overvoltage protection, a 3.6 V zener diode was connected in parallel to the ground at the output of the voltage regulator. When the output voltage rises above 3.6 V, the diode breaks down and conducts current significantly in reversed bias, thus preventing voltage from rising further. While LM317 is responsible for regulating voltage, zener diodes act as a secondary protection against incorrect resistor selection or abnormal voltage spikes due to faulty LM317. A zener diode was chosen as it clamps voltage effectively while providing a simple and low cost protection method.

4.2.7 Reverse Bias Protection

Requirement: When the voltage source is connected in incorrect polarity, the design must protect the output from being damaged due to reverse bias caused by reversed polarity.

Proposed Design: A 1N4007 power diode was selected and connected at the input of the voltage regulator. Under normal operation, the diode is forward-biased and allows current to flow into the LM317. Conversely, when the source polarity is reversed, the diode becomes reverse-biased and effectively blocks the current flow, preventing effective input voltage from

reaching the voltage regulator. This results in a zero output voltage. The diode was chosen as it offers stability while being cost effective, protecting components from high reverse current.

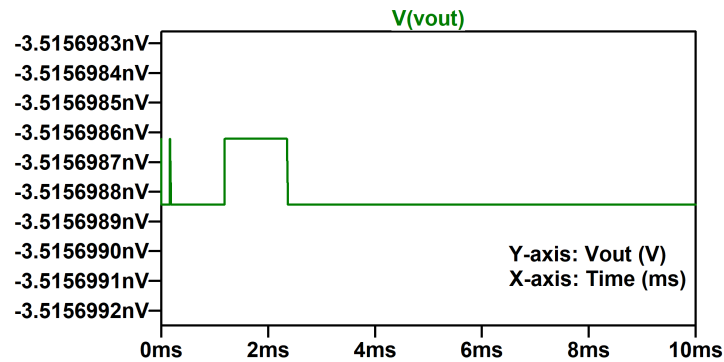


Figure 15: LT spice simulation when voltage source is connected in reverse direction

From Figure 15, the magnitude of output voltage is 3.516 nV, which is approximately zero. This successfully meets the requirement as the output is protected from reverse polarity.

Appendix B: Data and Measurements

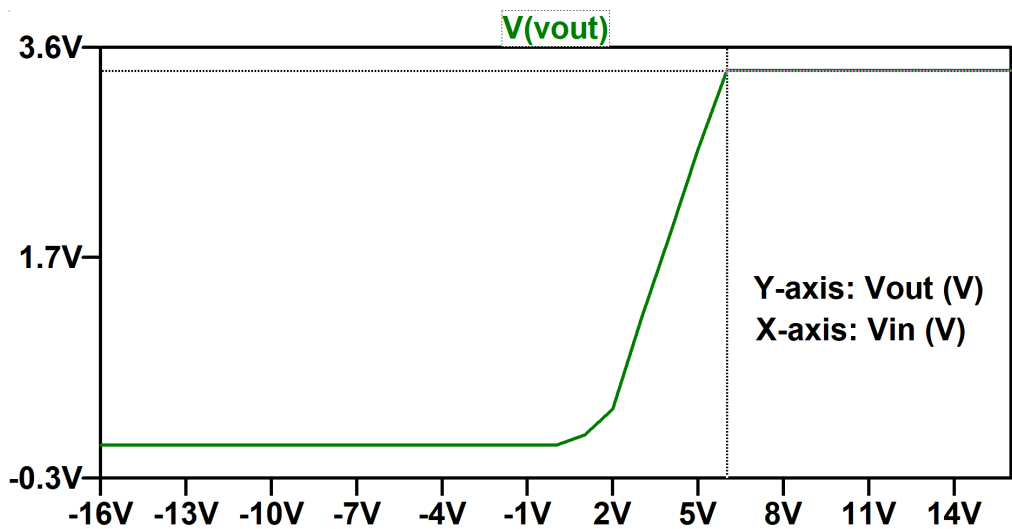


Figure B1: LT spice simulation result of output voltage using LM317 voltage regulator

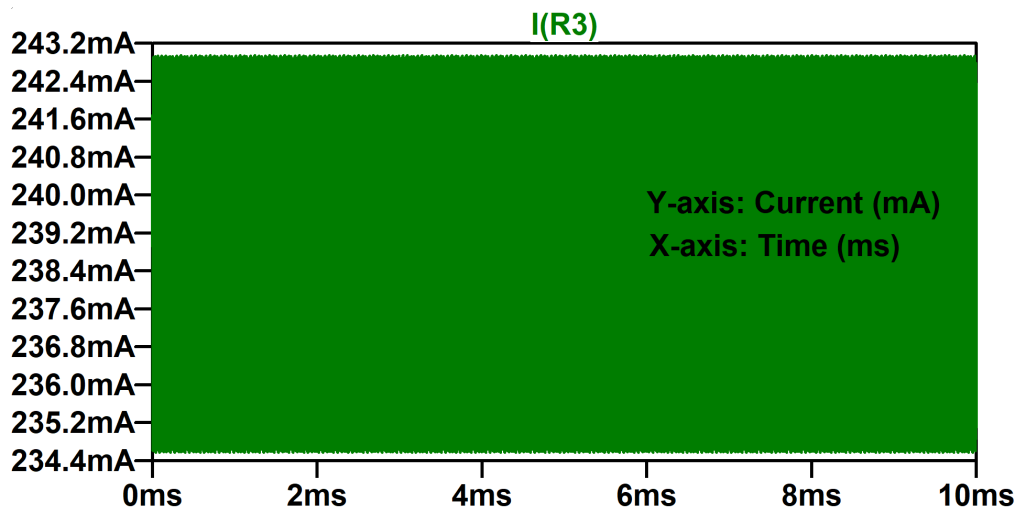


Figure B2: LT spice simulation result of DC load output current prior to ripple smoothing

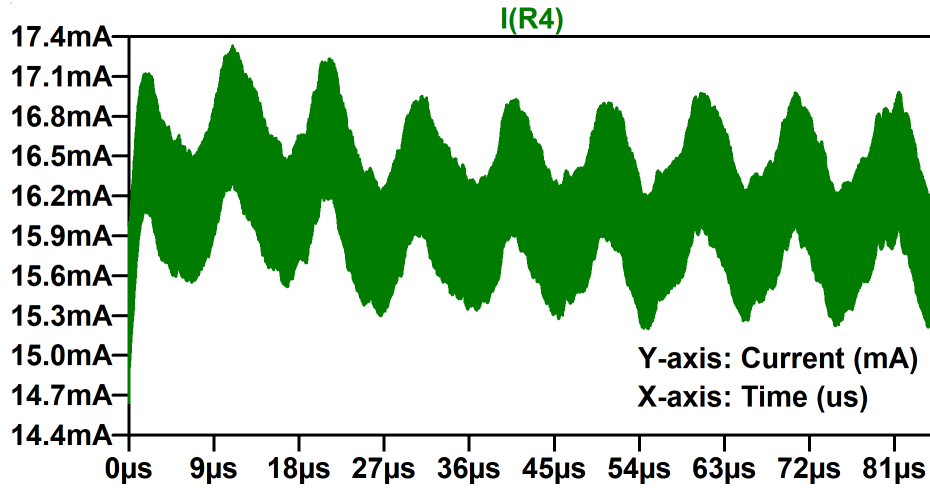


Figure B3: LT spice simulation result of active load output current prior to ripple smoothing

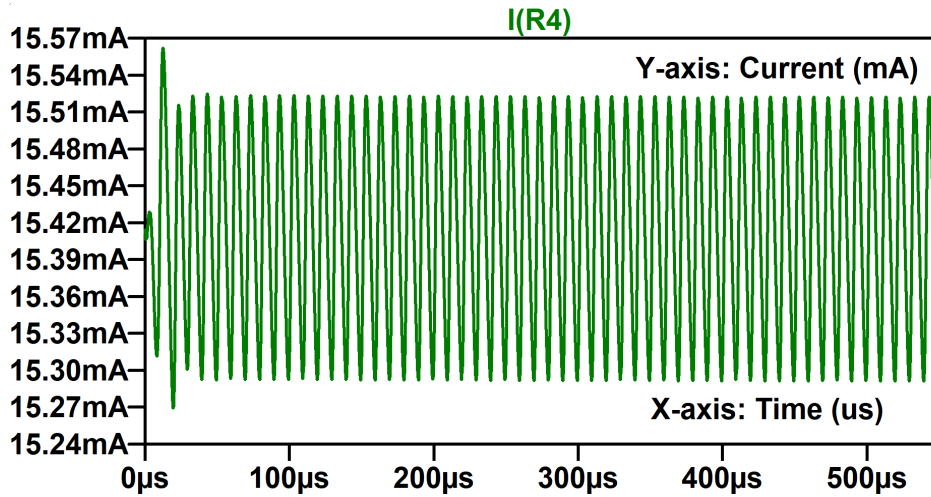


Figure B4: LT spice simulation result of current ripple under 80 MHz switching

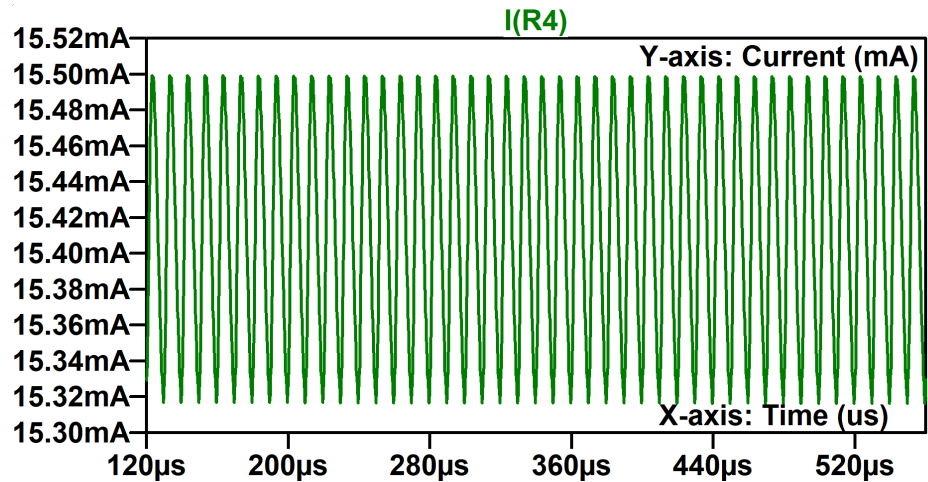


Figure B5: LT spice simulation result of current ripple under 1 MHz switching

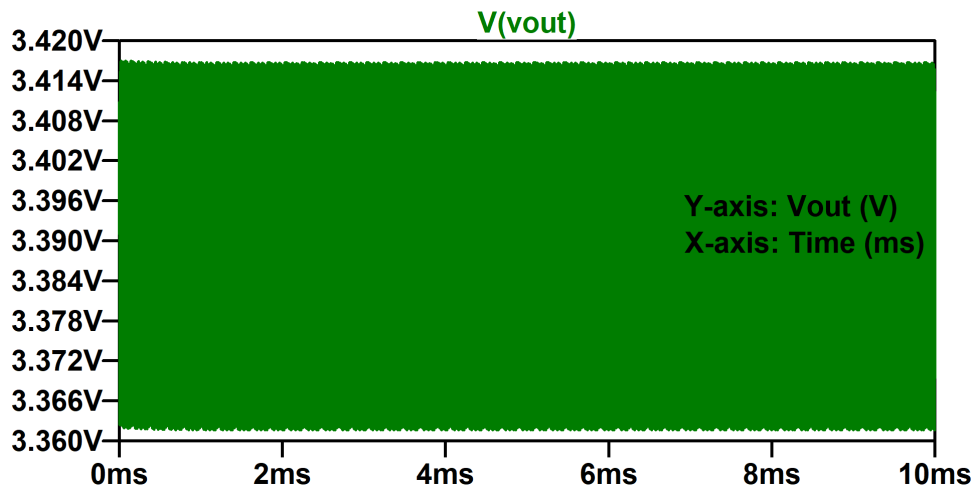


Figure B6: LT spice simulation result of voltage ripple under DC load

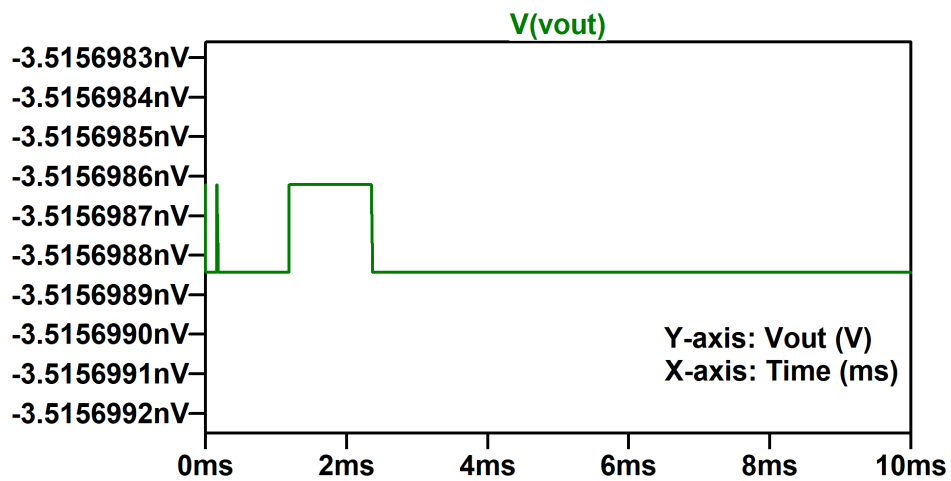


Figure B7: LT spice simulation result of output voltage when voltage source is connected in reversed polarity